

PROBoter - Automating PCB analysis tasks to support penetration tests of embedded systems

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Abstract. Analyzing the security of hardware components is an essential part of penetration tests in the context of embedded systems. Especially in the domain of automotive security and upcoming autonomous vehicles, vulnerabilities of electronic control units (ECUs) may have severe implications and therefore should be identified early in the development process (as also required by new regulations [7, 27]). Typically, the security analysis requires manual probing of pins or other contact pads on the printed circuit boards (PCBs) followed by the attempt to identify signals and respective bus protocols. To increase the efficiency of this hardware analysis and further reduce error-proneness of manual probing steps, we introduce the PROBoter (**PRO**Bing **ro**boter) as a novel open-source and self-calibrating architecture for automated PCB analysis as part of a penetration test. The PROBoter consists of four independently moving probes assisted by a camera for the detection of pins and contact pads, and an oscilloscope for signal measurement. Several algorithms allow the automatic probing of voltages at detected pins and contact pads, and subsequently try to identify existing bus protocols. We evaluated the feasibility of our approach using different PCBs and a self-designed test board for signal generation. Our evaluation shows that the PROBoter can detect and probe respective parts of a PCB with high accuracy and can successfully identify selected bus protocols. Our novel approach provides a significant contribution to the field of PCB analysis by automating manual work and thus increasing the efficiency of penetration tests of embedded systems such as automotive ECUs.¹

Keywords: Embedded Security · Hardware Security · Penetration Testing · Reverse Engineering · Automated Probe Testing.

1 Introduction

Security of embedded systems, such as IoT devices or automotive ECUs, has become increasingly relevant during the last decades. With their growing connectivity, previously isolated components are now exposed to remote attack attempts. Especially for upcoming autonomous vehicles, vulnerabilities can have severe implications that threaten the safety of drivers and other road users.

¹ This work is co-funded by the German Federal Ministry of Education and Research (BMBF) in the research project SecForCARS (<https://www.secforcars.de/>)

To identify potential vulnerabilities early in the development process, embedded systems can be subjected to one or more penetration tests. The security analysis of hardware components is an essential part of such a penetration test as direct physical access to bus communication or debug interfaces on the board are a common entry point for attackers. For instance, an existing UART interface on the board might allow access to an interactive shell of the underlying operating system run by a system on a chip (SOC). Another common attack vector is the exchange of key material via SPI or I²C bus protocols during boot time, e.g., to provide keys for a secure boot process.

Typically, the hardware security analysis includes manual probing of pins or other contact pads on the printed circuit boards (PCBs) followed by the attempt to identify signals and respective bus protocols. The last part is especially relevant if no data sheets are available for the analyzed components, as in black-box penetration tests. The whole manual probing process is often time-consuming and error-prone, as contact pads are often very small and manually placed probes can slip out of position or have insufficient electrical contact.

In order to increase the efficiency of this hardware analysis and further reduce errors due to the manual probing steps, we introduce the PROBoter as a novel open-source and self-calibrating architecture for automated PCB analysis.

The PROBoter consists of four independently moving probes. These are assisted by a camera for the detection of pins and contact pads. At the start of an analysis, the PROBoter creates a high resolution image of the complete PCB by moving the camera along a predefined path over the board and joining the partial images. From the complete image, potential position placements of the probes based on the detected pins and contact pads are calculated. The subsequent automatic probing process systematically measures the voltages of available signals for each contact point and attempts to identify used bus protocols such as UART, SPI, or I²C as potential entry points for an attacker.

We evaluated the feasibility of our approach using different PCBs and a self-designed test board for signal generation. The results show that the PROBoter can detect PCB components with a mean average precision of 87.4%, pins and contact pads with a success rate of up to 80.6% mean recall, achieves a probing granularity of $\pm 0.1\text{mm}$, and identifies bus protocols with a success rate of 98%. With our novel approach, we provide a significant contribution to the field of automated PCB analysis. This can support the identification of vulnerabilities early in the development process, which becomes more and more relevant especially in the automotive domain due to new regulations (e.g. ISO21434 [7] or UN R155 [27]). Here, the PROBoter may increase the efficiency of analyzing hardware components compared to the error-prone process of manual probing.

Outline: After giving an overview of related work in Section 2, we describe the details of the PROBoter in Section 3 followed by a discussion of the evaluation results in Section 4. We conclude our paper in Section 5.

2 Related Work

The functionality of the PROBoter covers three areas: automated PCB reverse engineering, visual PCB analysis, and automated protocol identification. The following paragraphs discuss related work in these areas.

Automated PCB reverse engineering: Carlson and Hauptmann [3] apply a Flying Probe Tester (FPT) in their work on reverse engineering of undocumented legacy PCBs. These types of machines are typically equipped with multiple independently movable electrical probes. In addition, cameras may be integrated for visual inspection. Carlson and Hauptmann use the built-in camera system of an FPT to reconstruct coordinates of possible probing points. In the next step, the FPT facilitates the automatic reconstruction of electrical connections by probing the previously identified locations.

FPTs are originally designed to automate inspection and probing as part of the quality assurance process during PCB manufacturing. They are used for this purpose since the 1980s [1]. Major drawbacks of industrial FPTs are their cost of usually more than 20k US dollar and their limited customization options for using them in different use cases, such as penetration tests. Here, customization is necessary as data like the PCB layout used as input to FPTs in the quality assurance process is usually unknown.

Ng [19] describes a similar process as proposed by Carlson and Hauptmann. He emphasizes the importance of software tools compatible with the FPT hardware which are often proprietary and closed-source products of the respective vendors.

Visual PCB analysis: The visual analysis of unknown PCBs can be separated into the fields of analyzing unpopulated and populated PCBs. The analysis of unpopulated PCBs (i.e., electrical components have been removed from the carrier board) aims at reconstructing the routing of electrical traces and connections. It is primarily based on using color information of the PCBs [17, 8, 25].

During a penetration test of embedded systems, usually populated PCBs are analyzed. This allows the analysis of the system's architecture in an unpowered state as well as the dynamic behavior in normal operation mode. When performed early in the development phase, the number of prototypes available for a penetration test is often limited. Therefore, the destructive steps required to analyze an unpopulated PCB are often not feasible. Instead, as much information as possible is gathered from the unmodified device. Kleber et al. [9] introduce an algorithm based on color segmentation to locate security-relevant components in images of populated PCBs. The focus of their work is the automatic gathering and processing of data sheets based on detected components.

Comparing and evaluating new methods for visual PCB analysis produces the need for standardized data sets. For this purpose, Pramerdorfer and Kampel [20] published the PCB-DSLR image data set containing 748 PCB images with 9,313 annotated integrated circuits (ICs) in total. Since then, further image data sets –both closed and freely accessible ones– have been created [16, 12, 15, 22].

Protocol identification: The main challenge for protocol identification is the identification of voltage characteristics of a measured signal. Burov [2] describes

a challenge similar to ours where electrical appliances are identified based on their power consumption patterns.

McClure and Such [18] provide a more application-oriented approach for identifying an automotive communication protocol. They present an adapter which tests the available pins and configures the communication accordingly.

If the protocol type is known, existing patents and algorithms can be applied to identify a protocol's baud rate. One method performs a combination of detecting frame errors and comparing expected and received characters [11]. Other approaches vary the decoding parameters until previously stipulated synchronization characters [4, 24] or valid frame structures are detected [10].

3 The PROBoter

The PROBoter architecture consists of several key components. First, the designed hardware platform and its auto-calibration routine are introduced in Section 3.1. Subsequently, adopted and newly developed algorithms for visual analysis of PCB image data in combination with two new PCB image data sets are presented in Section 3.2. Lastly, Sections 3.3 and 3.4 describe the techniques of the PROBoter for signal analysis and bus protocol identification.

3.1 Mechanics and Calibration

The central component of the novel PROBoter architecture is a modular hardware platform. Its design is inspired by the mechanical construction of commercial FPTs, but based on standard parts commonly found in state of the art 3D printers. This results in reduced production costs and facilitates replicas in the spirit of the open-source community. Based on the construction data, we assembled a first prototype as shown in Figure 1.

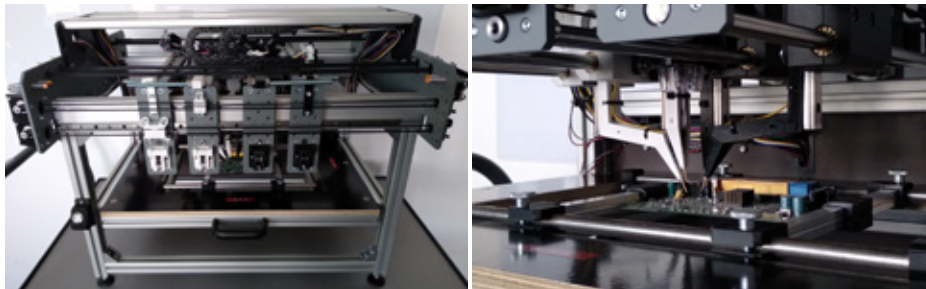


Fig. 1: Prototype of the PROBoter hardware platform.

Mechanical design: A frame built of standard aluminum profiles forms the base of the hardware platform. At the top, two profile rail guides are mounted that can carry up to four individual movable probing units. Each probing unit uses a set of three linear axes to allow an autonomous, three-dimensional movement of the electrical probe.

The PROBoter hardware contains a camera system for initial PCB image generation in form of a USB microscope attached to one of the probing units. By moving the camera and constantly capturing images, a final stitching process generates a high-resolution image of the PCB under test.

Control hard- and software: In contrast to the control software of existing FPTs, the PROBoter hardware is controlled by open-source software. The movement of a single probing unit is controlled by a 3D printer controller board running a modified version of the open-source 3D printer firmware *Marlin*².

To control the PROBoter platform as a whole, we designed and implemented a control software which will be published under an open-source license on GitHub³. The web-based software consists of a central server which controls the hardware platform. Additional micro services were implemented for visual analysis of PCB images and of measured voltage signals.

Calibration: Instead of using expensive measurement equipment commonly used for the calibration of high-precision FPTs, our approach is based on a low-cost 3D printed reference board shown in Figure 2. The developed calibration routine can compensate most of the linear errors resulting from imprecise part assembly or manufacturing tolerances of individual mechanical parts.

The reference board consists of electrically non-conductive plastic. Several metallic reference pins embedded at defined board locations, define a global Cartesian reference system S_{Glob} as shown in Figure 2. During the auto-calibration process, the centers of the reference pins are measured in the probing unit's (S_{Probe}) or camera system's (S_{Camera}) local coordinate systems. As the precise locations of these pins are known, the transformation matrix from a unit's local to the uniform global coordinate system can be calculated.

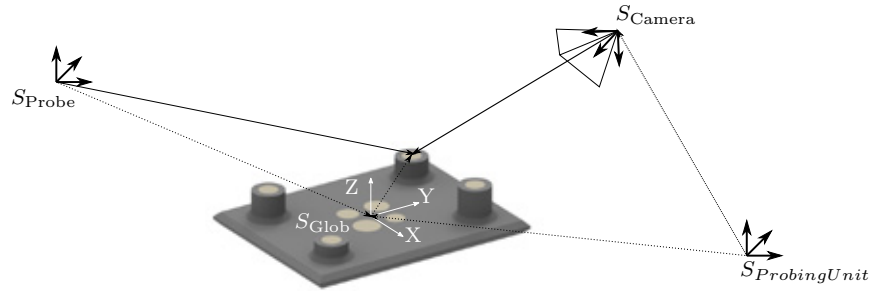


Fig. 2: Coordinate systems involved in the auto-calibration routine.

3.2 Component and Pin Detection

The PROBoter provides different strategies to automate the visual PCB analysis based on machine learning and classic computer vision algorithms. Existing

² <https://marlinfw.org/>

³ <https://github.com/schutzwerk/PROBoter>

PCB image data sets often target a single use case with a set of defined preconditions, like PCBs placed on a black conveyor belt as created by Pramerdorfer and Kampel[20]. As we designed the PROBoter as a generic platform for embedded system analysis, we also present three new image data sets (two for PCBs and one for ICs) that show a big variety in the visual appearance of PCBs. These data sets can be used stand-alone or in conjunction with existing PCB image data to evaluate algorithms for populated PCB analysis.

Image data sets: To test the robustness of the identification and localization of active components mounted on populated PCBs, we created an image data set based on results of the Google image search *microcontroller*, *pcb*, *pcb red* and *platine* (the German word for PCB). The resulting data set PCB-GOOGLE consists of 190 images of populated PCBs with 1,100 manually labeled IC object instances. The images in the set show a wide variety of PCBs regarding solder mask color as their component package types and zoom levels range from macro views of single ICs to arrangements of multiple PCBs.

To represent PCBs used in state of the art embedded systems, we created a second data set PCB-INTERNAL from internal PCB sources. It consists of 15 unique PCB images and 53 annotated IC instances.

The third proposed image data set, IC-PINS, aims at the evaluation of methods used for the precise localization of pins connecting an IC to the underlying PCB solder pads. It is extracted from the data set PCB-INTERNAL and contains 46 IC macro images with 1,181 annotated component pins. The number of pins per image range from zero visible pins for Ball Grid Array (BGA) packaged ICs up to 100 typically found at Thin Quad Flat Packages (TQFP).

Component detection: During the visual analysis of a populated PCB, the first step is the identification and localization of security-relevant components. In recent years, great progress has been made in this area of object detection by the development of (convolutional) neural networks [29][28]. For the PROBoter platform, the two well-proven object detectors *Faster R-CNN* [21] and *SSD* [14] were trained on the existing PCB-DSLR image data published by Pramerdorfer and Kampel [20] and the newly created PCB-GOOGLE data set. The TensorFlow object detection API [6] was selected as base framework. It provides enhanced implementations of the original neural network architectures. In addition, pre-trained models on the COCO image data set [13] are provided in the object detection API's model zoo⁴. These models were used for transfer learning of the single object class IC contained in the PCB image data sets.

The PCB-INTERNAL data set was used only in the evaluation phase to compare different object detectors and data sets together with their combinations.

Pin detection: Besides classic test pads, the pins connecting an IC to the underlying PCB are ideal candidates for probing during an embedded system analysis. If a pinout of the IC is found in a data sheet, the pin's functionality is known and does not need to be reverse engineered.

⁴ https://github.com/tensorflow/models/blob/master/research/object_detection/g3doc/tf1_detection_zoo.md

To automate the task of pin localization and identification, we propose two different algorithms. The input for both algorithms is an image showing a single IC with a slight surrounding padding area. The first approach uses the same object detectors already used for the initial component identification in populated PCB images. The detectors have been trained on a subset of the proposed IC-PINS data set. From the rectangular object detection results, the pin center is defined as the center of the resulting bounding box.

In addition, we propose a second algorithm based on classic computer vision with a multi-stage computation pipeline. In a first step, we use flood-filling to remove the solder mask and plastic areas covering the IC. By applying morphological operations and skeletonization techniques in a second step, the individual pins are isolated further. As a last step, the center of the remaining connected components is calculated which represent the final pin candidates.

3.3 Voltage Record Analysis

After the PROBoter has finished the more hardware focused analysis of the PCB, in the consecutive step it measures and evaluates the voltages occurring on the PCB. This results in the identification of the underlying protocol of the signal. This knowledge is fundamental for attacking the communication between the components on the board. Currently, the scope of detectable protocols is limited to 1-Wire, I²C, SPI, and UART for having a first proof of concept.

The evaluation approach is based on a combination of a statistical *evaluation of measured voltages*, an *evaluation of logical state durations*, a *delta analysis process*, and an *autocorrelation method* as described in the following paragraphs.

Evaluation of voltages: For each signal, two properties are extracted from a histogram of occurring voltage values: the amount and levels of the most prominent voltages, and the widths of the peaks. The latter ones support the distinction of transitions between a signal's voltage states.

Evaluation of logical state durations: To evaluate the durations of logical states, the PROBoter initially identifies the moments of distinct voltage level changes and determines the durations between each change. To distinguish the

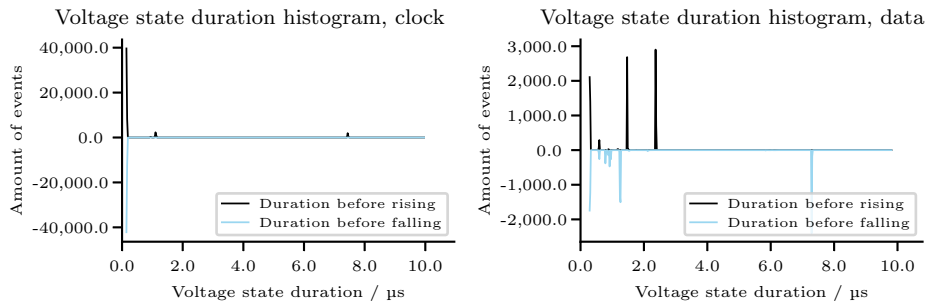


Fig. 3: Statistical evaluation of voltage state durations by histograms for an exemplary clock signal (left) and an exemplary data signal (right).

durations of lower and upper voltage states, the algorithm plots the states on the vertical axis of a histogram (see Figure 3 for plots based on exemplary clock and data signal records). The positive values describe the amount of events for which the signal state rises, while the negative values describe the amount of events for which the signal state drops.

Delta analysis: A Delta analysis allows the identification of transmission blocks by observing the minimum and maximum amount of voltage state changes within an increasing time interval [26]. This method leads to a staircase shaped plot (see Figure 4) with increasing duration of the interval on the x-axis and the amount of observed events on the y-axis. Reoccurring intervals of inactivity appear as steps of prolonged durations.

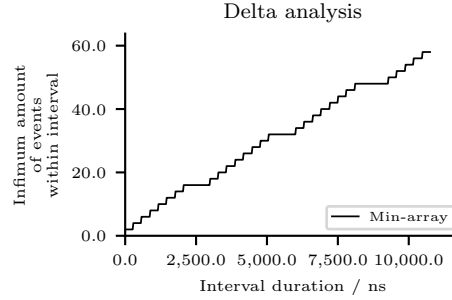


Fig. 4: Delta analysis showing signal state changes over time.

Autocorrelation method: The autocorrelation function allows the identification of repeating patterns. Such a pattern occurs for the UART protocol as sequence of the stop and start bit [23]. In combination with the evaluation of the duration histograms, the result of this evaluation is the basis for the identification of UART transmission parameters.

3.4 Signal and Protocol Identification

Based on the described analysis methods, the PROBoter attempts to identify the following properties from a voltage record: the *line function*, the *underlying protocol*, the *association of data signals to clock signals*, and the *transmission parameters for the UART protocol*.

Identification of line function: To identify the function of a measured line, i.e., power supply, ground, clock, data or communication control, the PROBoter extracts characteristic traits from the record. The algorithm observes the amount of activity and variation of voltages which is already evaluated for the generation of the duration histograms.

Identification of protocol: If the PROBoter identifies a clock or data line, it continues the evaluation of the data record. Therefore, the algorithm compares the characteristics it evaluated for a data record with expected characteristics

from a list of protocols. The protocol which has the highest amount of matching characteristics is assumed to be the correct protocol type.

Association of data signals to clock signals: For synchronous transmissions, the PROBoter attempts to associate the clock signal records with one or more matching data signal records. For this linking process, the algorithm leverages the fact that time intervals of clock line activities are related to the time intervals of data line activities. Furthermore, the duration of a clock cycle is similar to the duration of a symbol on the matching data line. The more similar the activity intervals and symbol durations are for a clock and a data line, the higher the chance that both lines are used for a common synchronous transmission.

Identification of UART parameters: If the UART protocol is detected, the PROBoter automatically tries to identify the respective transmission parameters, i.e., baud rate, amount of data bits per frame, amount of stop bits, and the usage of parity bits. This information is combined with the duration results from the autocorrelation method.

4 Evaluation

In order to evaluate the performance of the PROBoter in a real-world scenario, the individual components have been analyzed in detail. We first discuss the results of the hardware platform evaluation including measurements of general mechanical characteristics for the probing accuracy. We then present the component and pin detection results achieved with our newly introduced PCB and pin image data sets. The chapter concludes with the evaluation results of our developed algorithms for voltage signal analysis and protocol identification.

4.1 Probing Accuracy

The PROBoter hardware platform has been evaluated for repetition and absolute positioning accuracy as these two properties are most essential for a reliable usage in a real-world assessment of an embedded system.

For the evaluation, we repeatedly measured the pins of the reference board with the movable camera system and all electrical probing units individually. For each pin measurement, we recorded the probed center coordinate. In the next step, we calculated the errors to the well-known reference pin coordinates in the common global coordinate system. For the transformation into the common coordinate system, we used the matrices determined during a previously performed component calibration as described in Section 3.1. The measurements have been performed 10 times for each component. The results are shown in Table 1.

Component	max. error range X	max. error range Y	max. error range Z
	in mm	in mm	in mm
Camera system	0.140	0.130	0.180
Probing unit 1.1	0.069	0.056	0.296
Probing unit 1	0.081	0.059	0.342
Probing unit 2	0.070	0.097	0.423
Probing unit 2.1	0.053	0.059	0.269

Table 1: Probing accuracy results of the PROBoter hardware platform.

The maximum error range of all probing units is below 0.1mm and therefore allows the reliable probing of test pads and IC pins. The larger error range in the vertical Z-axis results from inaccuracies in the assembly of the reference board. However, the used spring-loaded electrical probes can compensate height differences in the probing plane. The repeatability of the movable camera system is with a maximum error range of 0.140mm (Y direction) not as accurate as the one of the electrical probing units. This might be a result of the limited absolute resolution of the currently used camera.

To evaluate the PROBoter hardware with a real PCB, we designed a special evaluation board as a blank PCB with unpopulated solder pads. The pads simulate possible probing points that can be found on typical state of the art PCBs ranging from 0.53 to 0.2mm in pad and 0.35 to 0.67mm grid size. Like in a real embedded system analysis, we first scanned the board with the PROBoter's camera system. We then selected the test pads manually in the final image and probed all pads with each probing unit individually. For 10 subsequent evaluation runs, the PROBoter successfully probed all test pads on the evaluation board, proving the feasibility in a real-world scenario.

4.2 Component and Pin Detection Accuracy

The proposed algorithms for component and pin detection have been evaluated independently. For the pin detection, we introduce a new metric to reflect the requirements of automated electrical probing with the PROBoter hardware.

Component detection results: We trained the selected neural networks on the existing PCB-DSLR and our proposed PCB-GOOGLE data set. We used a training to evaluation split of 4:1 to monitor the training progress. In addition to the original images of the data sets, we further extended the limited number of existing image data by applying standard augmentation operations built into the TensorFlow object detection API input pipelines. We trained all models until train loss convergence. This was reached for all combinations of model and image data sets after at most 40,000 training iterations.

To compare the different model architectures and the influence of different PCB image data sets, we evaluated the trained models on our PCB-INTERNAL data set with the Pascal VOC metric [5]. The results are summarized in Table 2. Both models trained on our PCB-GOOGLE data set provide good detection results which is reflected by high mAP scores and the precision-recall curve shown in Figure 5. Independent of the architecture, models trained on the PCB-DSLR image data set show remarkably lower detection accuracy.

This proves that our approach of enforcing generic PCB analysis without any preconditions is appropriate for this use case.

Model	Image data set(s)	mAP in %
SSD	PCB-DSLR	42.11
SSD	PCB-GOOGLE	81.52
Faster R-CNN	PCB-DSLR	57.11
Faster R-CNN	PCB-GOOGLE	87.38
Faster R-CNN	PCB-DSLR + PCB-GOOGLE	15.89

Table 2: mAP of the trained models for component detection.

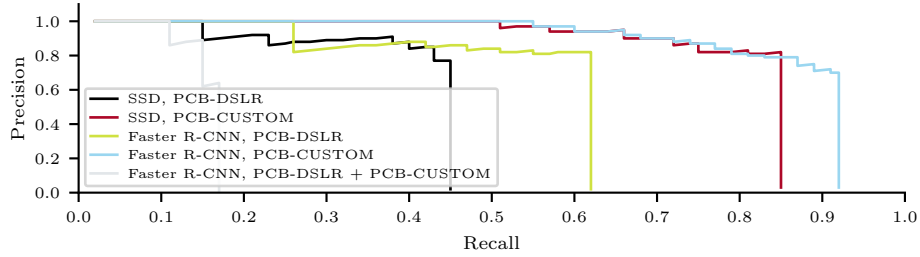


Fig. 5: Precision-recall curves of the trained models for component detection.

Pin detection results: Till now, there is no publicly available image data set containing labeled IC pins. Therefore, we trained and evaluated our proposed pin detection algorithms solely on our IC-PINS data set. Due to insufficient image quality, we removed eight images resulting in a data set with 38 IC images.

For our approach based on neural networks, we used the same model configurations and augmentation operations as for component detection. We used again a 4:1 training to evaluation split, this time of the IC-PINS data set for model training. The parameters of our proposed computer vision pipeline were manually fine-tuned to produce the best results on the IC-PINS data set.

To evaluate our proposed method for IC pin detection, we used the following three metrics: Precision and recall values, inference time, and a newly defined metric named Relative Border Distance (RBD). For all predicted pin detections in the algorithm's output, we calculated the averaged precision and recall values mP and mR according to the Pascal VOC metric. For outputs of the computer vision pipeline, a pin detection is rated as True Positive (TP) if the pin center is inside a labeled pin area. If multiple detections match these criteria, all further pins are marked as False Positive (FP). The results of the pipeline are not sorted as this is done by the Pascal VOC metric due to the lack of confidence values. For a successful automated electrical probing, it is required to detect the center of the pin that connects it to the PCB's solder pad. Therefore, we introduce the new metric of Relative Border Distance (RBD) and the averaged value over multiple pin detections mean Relative Border Distance (mRBD). For each TP of the pin proposals, the relative distance of the predicted pin center to the borders of the corresponding Ground Truth (GT) pin area is calculated in both, X and Y image coordinates. The resulting relative border distances are multiplied in the next step to calculate the final RBD value in the range of [0,1].

Table 3 gives an overview of the pin detection results. Our proposed computer vision pipeline as well as the trained Faster R-CNN model show a high pin detection rate. Both algorithms produce only few FPs which is reflected by high mR scores. In comparison, the SSD model shows a significantly reduced detection rate and also produces more FPs. Regarding the prediction of the pin center coordinates, the Faster R-CNN model produces the best results followed by the CV pipeline. Due to the fact that all algorithms have been optimized either manually or by numerical optimization during model training, further

Algorithm / model	mP in %	mR in %	mRBD	Avg. t_{inf} in sec.
CV pipeline	72.5	80.6	0.441	0.350
Faster R-CNN	86.3	77.9	0.614	29.314
SSD	55.5	21.4	0.163	2.695

Table 3: Overview of the pin detection evaluation results.

trials are already planned to validate the robustness and ability to transfer the results on new image data.

4.3 Accuracy of Signal and Protocol Identification

In order to evaluate the accuracy of the various protocol identification approaches, we used a self-designed test board and a USB to UART converter for signal generation. For all protocols, the generated signals are based on the transmission of random messages. The signals differ in length, binary entropy and transmission speed of the transmitted message.

Accuracy of protocol identification: For the identification of the used protocol (1-Wire, I²C, SPI, or UART), we generated 54 samples with variations in length of the transmitted message, transmission speed and message content. The samples were split into groups of 9 samples of the same content and length. Finally, we evaluated the algorithm’s identification rate of the used protocols for each group of samples. For 53 of the 54 samples, the algorithm correctly identified the used protocol. For one UART protocol sample, the algorithm misidentified the transmission as an additional data line of an SPI transmission. In both samples, the algorithm determined similar durations for the activity intervals of both sample records as well as for the UART symbol and the SPI clock cycle (see Figure 6 for a graphical representation). For each protocol sample, the algorithm produces a value how well it discerns the correct protocol. Due to the combinations of various speeds and encoding types, this results in a range of distinction ratings. The higher the rating, the better the distinction. The negative rating for the outlier of the UART protocol is caused by the false attribution of the sample to an SPI transmission.

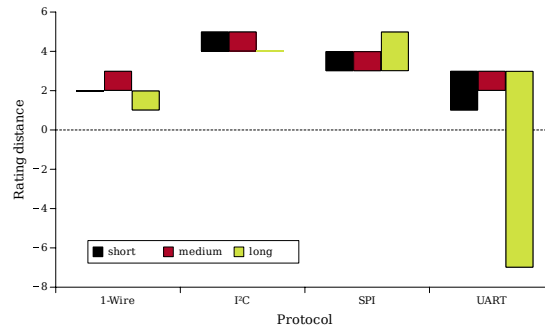


Fig. 6: Success rating to differ protocols for samples with different length.

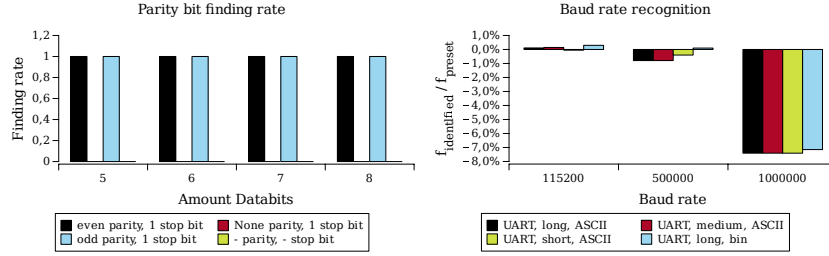


Fig. 7: Finding ratings for UART transmission parameters of generated samples.

Accuracy of UART transmission parameter identification: To evaluate the identification rating of UART transmission parameters, 34 samples were generated with variations in the baud rate, amount of data bits per frame, usage of parity bits, and amount of stop bits. The transmission parameters were correctly identified for all of these 34 samples (see Figure 7). The PROBoter correctly identified the amount of data bits, usages of parity bits, amount of stop bits and baud rate. In case of the baud rate, the deviation for $f = 1\text{Mbaud}$ is caused by an inaccuracy of the signal generator.

Accuracy of correlation between data and clock signals: To determine the rate of correct linking between clock and data lines of synchronous transmissions, 24 sample sets consisting of data and clock signal lines were generated with offsets of 1, 10 and 100 μs .

The PROBoter correctly linked the signals of the data and clock lines based on their original transmissions for all 24 samples. To generate a high similarity, the transmissions only differ by the offsets of start and end of transmissions. For the quality rating, the product of the similarity of symbol durations and signal activity is plotted on the Y-axis. Between two signals, the similarity of duration of activity and duration for symbols can range from 0 (worst) to 1 (best) each. Figure 8 plots the signal similarities of the evaluated samples.

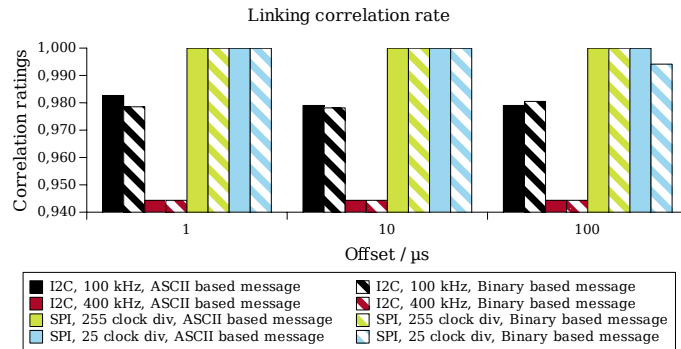


Fig. 8: Quality result of linking samples of synchronous transmissions.

5 Conclusion

We introduced the PROBoter as a novel open-source and self-calibrating architecture for automated PCB analysis as part of a penetration test. Our evaluation showed that the four independently moving probes of the PROBoter allow a precise and systematic measurement of available signals for each pin or contact pad. Furthermore, the signal detection algorithms effectively identified existing bus protocols (evaluated for 1-Wire, UART, SPI, and I²C). This generates potentially interesting entry points for an attacker.

The PROBoter provides a significant contribution to the field of automated PCB analysis, especially when conducted as part of penetration tests. Compared to the error-prone alternative of manual probing, the PROBoter can increase the efficiency of analyzing hardware components and further increase the likelihood of discovering vulnerabilities early in the development process. Especially in the automotive domain this becomes more and more relevant due to new regulations, such as ISO21434 [7] or UN R155 [27].

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